

A Polynomial Passive-Dose Lower Bound for Fourfold Forrelation

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Abstract

We give a standalone derivation of a polynomial lower bound for deciding the exact two-sided fourfold-forrelation promise by adaptive fresh passive optical batches. A batch may use an arbitrary mixed bosonic state whose sign-mode photon number is at most its charge on the entire support, arbitrary bypass idlers, one direct-sum sign-mask interaction, and an arbitrary collective measurement. Between batches only a classical outcome is retained. For every power of two $N \geq 2^{30}$, worst-case error at most $1/3$ implies

$$D \geq \frac{2}{15} N^{1/8}.$$

The exponent is a fixed unconditional positive constant; it does not meet the original stronger target $\alpha > 1/4$. The argument uses a fresh-pivot inequality that controls arbitrary fixed-level Fourier functionals of the full adaptive protocol, including all cross-batch symmetric-difference cancellations. A correlated Gaussian sign construction then supplies signed Fourier moments of order $N^{-3r/8}$ at level r , while a separate concentration argument moves the average-case construction to the exact promises $\text{Forr}_4 \geq 1/4$ and $\text{Forr}_4 \leq -1/4$.

Verification status. This derivation was produced with substantial large-language-model assistance. Its definitions, constant ledger, and compilation have been internally audited, but it has not received independent end-to-end review by a subject-matter expert. It is therefore a provisional companion result, matching the status stated in the main paper. In particular, Theorem 4.1 and its fresh-cut construction are the principal load-bearing claims for independent verification. The label “unconditional” below refers only to the absence of a conjectural mathematical hypothesis, not to the epistemic status of this package.

1 Problem statement

Fix an integer $d \geq 1$, put $N = 2^d$, and identify $[N]$ with $\mathbb{F}_2^d$. Let

$$H_{ij} = N^{-1/2}(-1)^{\langle i,j \rangle}, \quad |u\rangle = N^{-1/2} \sum_i |i\rangle.$$

For

$$x = (x^{(1)}, x^{(2)}, x^{(3)}, x^{(4)}), \quad x^{(b)} \in \{\pm 1\}^N,$$

write $D_b = \text{diag}(x^{(b)})$ and define

$$\text{Forr}_4(x) = \langle u | D_1 H D_2 H D_3 H D_4 | u \rangle.$$

Equivalently, with $h_{ij} = (-1)^{\langle i,j \rangle}$,

$$\text{Forr}_4(x) = N^{-5/2} \sum_{i,j,k,l} h_{ij} h_{jk} h_{kl} x_i^{(1)} x_j^{(2)} x_k^{(3)} x_l^{(4)}.$$

The exact promises are

$$\Pi_+ = \{x : \text{Forr}_4(x) \geq 1/4\}, \quad \Pi_- = \{x : \text{Forr}_4(x) \leq -1/4\}.$$

A passive batch of charge $t \in \mathbb{Z}_{\geq 0}$ prepares an arbitrary, possibly mixed, bosonic state on the $4N$ sign modes and arbitrary bypass idlers, supported on the sector in which the total sign-mode photon number is at most t . It applies exactly once

$$\Gamma((D_1 \oplus D_2 \oplus D_3 \oplus D_4) \oplus I_{\text{idler}})$$

and then performs an arbitrary collective POVM with a finite or standard-Borel outcome space. Number superpositions, vacuum coherence, repeated mode occupation, bosonic symmetrization, arbitrary idlers, heralding, singular outcomes, and postselection are allowed; the charge is the maximum sign-mode photon number on the support, not its expectation.

A passive protocol is an arbitrary adaptive tree of fresh batches. Only the classical outcome and private classical randomness cross a node boundary; no quantum state or quantum memory does. The next state, charge, POVM, stopping rule, and output may depend on the full classical history. If the charge at history h is t_h , the hard dose is

$$D = \sup_{\text{maximal branches } \gamma} \sum_{h \in \gamma} t_h,$$

where the sum on an infinite branch means the supremum over its finite prefixes. Whenever $D < \infty$, it is a nonnegative integer because it is a bounded supremum of integer-valued branch-prefix charges. Every traversal is charged, including failed, heralded, controlled, inverse, or conditioned traversals; conditioning never discounts dose. A zero-charge continuation is input-independent and may be absorbed into classical randomness, with nontermination treated as a terminal outcome. The protocol decides the promise with error at most $1/3$ when it outputs the correct sign with probability at least $2/3$ for every individual input in $\Pi_+ \cup \Pi_-$.

Theorem 1.1 (Main theorem). *Let*

$$\alpha = \frac{1}{8}, \quad c_0 = \frac{2}{15}, \quad N_0 = 2^{30}.$$

For every power of two $N \geq N_0$, every passive protocol in the full model above that decides Π_+ versus Π_- with worst-case error at most $1/3$ satisfies

$$D \geq c_0 N^\alpha.$$

2 Walsh notation and the bosonic slot embedding

Put $M = 4N$ and identify the sign coordinates with $[M]$. Thus N is the number of coordinates on each of the four masks, while M is only temporary notation for the total number of Boolean sign coordinates. For $Q \subseteq [M]$, write

$$\chi_Q(x) = \prod_{a \in Q} x_a, \quad \widehat{f}(Q) = 2^{-M} \sum_{x \in \{\pm 1\}^M} f(x) \chi_Q(x).$$

2.1 The exact labeled-slot embedding

Let

$$\mathcal{Q} = \mathbb{C}|\perp\rangle \oplus \mathbb{C}^M,$$

where $|\perp\rangle$ is a vacuum symbol, and define

$$O_x|\perp\rangle = |\perp\rangle, \quad O_x|a\rangle = x_a|a\rangle \quad (a \in [M]).$$

For every t , the truncated sign-mode Fock space

$$\bigoplus_{n=0}^t \text{Sym}^n(\mathbb{C}^M)$$

embeds isometrically into $\mathcal{Q}^{\otimes t}$: an n -photon symmetric vector is mapped to the symmetric t -slot vector with $t - n$ vacuum symbols. Under this embedding, the second-quantized sign mask is exactly $O_x^{\otimes t}$. Hence a charge- t batch is an arbitrary t -slot parallel phase query with arbitrary ancilla.

For $\mathbf{a} = (a_1, \dots, a_t) \in (\{\perp\} \cup [M])^t$, define

$$\partial\mathbf{a} = \{a \in [M] : \#\{s : a_s = a\} \text{ is odd}\}.$$

Then

$$O_x^{\otimes t}|\mathbf{a}\rangle = \chi_{\partial\mathbf{a}}(x)|\mathbf{a}\rangle.$$

This representation retains repeated occupation, bosonic symmetrization, number superpositions, and vacuum coherence. An arbitrary purification and every bypass idler are included in an additional ancilla factor.

3 Finite fresh trees

We first treat a finite tree with finite outcome sets. Purify every prepared state. At a nonterminal history h , let t_h be the charge, let

$$|\psi_h\rangle \in \mathcal{Q}^{\otimes t_h} \otimes \mathcal{A}_h$$

be the prepared state, and let $\{M_{h,y} : y \in Y_h\}$ be the POVM. Use the square-root instrument

$$W_{h,y} = M_{h,y}^{1/2}, \quad \sum_{y \in Y_h} W_{h,y}^* W_{h,y} = I.$$

At a terminal history h , let $a_h \in [0, 1]$ be the conditional probability of outputting $+$.

Define recursively a vector-valued accepted amplitude. At a terminal node, put

$$\Phi_h(x) = \sqrt{a_h} \in \mathbb{C}.$$

At a nonterminal node, put

$$\Phi_h(x) = \bigoplus_{y \in Y_h} (W_{h,y} O_x^{\otimes t_h} |\psi_h\rangle) \otimes \Phi_{hy}(x). \quad (1)$$

The tensor product in (1) records the fresh boundary: the child protocol acts on a newly prepared system and never acts on the residual output system of the parent batch. In the pivot proof this

is used both to construct a genuine separating cut through all marked parallel slots and to factor every continuation as an operator on a fresh tensor factor. Directly,

$$\|\Phi_h(x)\|^2 = \sum_y p_h(y|x) \|\Phi_{hy}(x)\|^2.$$

Thus, at the root,

$$f(x) = \|\Phi_\emptyset(x)\|^2, \quad (2)$$

where $f(x)$ is the probability of outputting x .

Along a branch, the total number of query slots is at most D . Consequently $\Phi_\emptyset$ is a vector-valued Walsh polynomial of degree at most D , and

$$\hat{f}(Q) = 0 \quad (|Q| > 2D). \quad (3)$$

4 The fresh-pivot inequality

The next theorem is the structural core of the proof.

Theorem 4.1 (Fresh-pivot inequality). *Let f be the acceptance probability of a passive protocol of hard dose at most D . If $1 \leq r \leq 2D$ and $\{c_Q : Q \subseteq [M], |Q| = r\}$ is any family of complex numbers, then*

$$\left| \sum_{|Q|=r} c_Q \hat{f}(Q) \right| \leq \left(\sum_{k=0}^r \binom{D}{k} \binom{D}{r-k} M^{\min(k, r-k)/2} \right) \max_{|Q|=r} |c_Q| \leq \binom{2D}{r} M^{r/4} \max_{|Q|=r} |c_Q|. \quad (4)$$

The estimate holds for arbitrary standard-Borel outcomes, singular histories, unrestricted tree width, private randomness, and infinite zero-charge tails.

4.1 Last-occurrence records

Expanding (1) in all labeled slot bases gives

$$\Phi_\emptyset(x) = \sum_\gamma \chi_{\partial\gamma}(x) v_\gamma,$$

where γ ranges over complete ket execution histories, including the classical terminal path and every slot word, and $\partial\gamma$ is the symmetric difference of the slot parities along that history. Therefore

$$\hat{f}(Q) = \sum_{\substack{\gamma, \gamma' \\ \partial\gamma \Delta \partial\gamma' = Q}} \langle v_{\gamma'}, v_\gamma \rangle. \quad (5)$$

The direct-sum outcome registers make the inner product zero unless the two histories have the same classical terminal path, while the tensor products retain the fresh factorization along that path.

Pad every ket execution and every bra execution by vacuum slots so that each has exactly D slot positions, ordered chronologically within its own side.

A doubled slot history contributes to $\hat{f}(Q)$ exactly when the ket and bra multiplicities have different parity precisely on Q . Thus, for every $a \in Q$, exactly one of the two sides contains a an

odd number of times. On that odd-parity side, mark the last occurrence of a . These $r = |Q|$ marked positions are distinct. Their set

$$R \subseteq (\{\text{ket}\} \times [D]) \sqcup (\{\text{bra}\} \times [D]), \quad |R| = r,$$

is the parity-side last-occurrence record. Every surviving doubled history has one and only one record, so the number of possible records is at most $\binom{2D}{r}$ (empty record classes do not matter).

For a fixed record, let $L \subseteq R$ be the marked ket positions and let $R \setminus L$ be the marked bra positions. If $k = |L|$, the coordinate values at the marked ket positions form a tuple $\alpha \in [M]^k$, and those at the marked bra positions form $\beta \in [M]^{r-k}$. A combined tuple (α, β) with any repeated coordinate value is declared inadmissible.

4.2 Parity tracking and exact record filters

Let

$$\mathcal{K} = \ell_2(\mathcal{P}([M]))$$

with basis $|S\rangle$, $S \subseteq [M]$. For $T \subseteq [M]$, let

$$X_T|S\rangle = |S \Delta T\rangle.$$

At every unmarked query slot, the coordinate value toggles the parity register. At a marked slot, the value is exposed as an open coordinate index and its toggle is omitted. If the exposed ket tuple is $\alpha = (\alpha_1, \dots, \alpha_k)$, write

$$A(\alpha) = \{\alpha_1, \dots, \alpha_k\}, \quad \Pi_\alpha^{\text{res}} = \sum_{S \cap A(\alpha) = \emptyset} |S\rangle\langle S|,$$

and define $B(\beta)$ and Π_β^{res} analogously. The ket side is projected at the terminal cut by Π_α^{res} , the bra side by Π_β^{res} , and the two residual parity registers are then identified by the identity map on $\mathcal{K}$.

The last-occurrence condition is imposed separately on each side. Once a coordinate has been exposed at a marked slot, every later slot on that same side is projected away from that coordinate. These restrictions and the terminal residual-parity restrictions are controlled orthogonal projections internal to one side. No row-index-controlled projection acts on the bra side, and no column-index-controlled projection acts on the ket side.

These filters are equivalent to the record definition. Indeed, for an exposed ket coordinate $a \in A(\alpha)$, omission of the marked occurrence and the projection Π_α^{res} make the remaining ket multiplicity of a even. Hence the full ket multiplicity is odd. Equality of the two residual parity sets makes the bra multiplicity even. The argument for an exposed bra coordinate is symmetric. Every unexposed coordinate has the same parity on the two sides. The no-later-occurrence projections make the designated occurrence the last one on its odd-parity side. Consequently every surviving doubled history is admitted by exactly its true parity-side last-occurrence record and by no other record.

Lemma 4.2 (Fresh cut contraction). *Fix a record R and its ket/bra split. The total contribution of all doubled histories with this record can be written*

$$\sum_{\alpha, \beta} C_{\alpha\beta} \langle u_\alpha, E_{\alpha\beta} v_\beta \rangle, \tag{6}$$

where

$$\sum_{\alpha} \|u_\alpha\|^2 \leq 1, \quad \sum_{\beta} \|v_\beta\|^2 \leq 1,$$

and the block operator

$$E = [E_{\alpha\beta}] : \bigoplus_{\beta} \mathcal{V}_{\beta} \longrightarrow \bigoplus_{\alpha} \mathcal{U}_{\alpha}$$

satisfies $\|E\| \leq 1$. The scalar entry is

$$C_{\alpha\beta} = \begin{cases} c_Q, & \text{if all entries of } (\alpha, \beta) \text{ are distinct and form } Q, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. We construct the vectors and the operator by cutting the finite doubled execution network at the marked coordinate resolutions.

For a slot q , let $P_a^{(q)}$ be the coordinate projector on that slot, with $a \in \{\perp\} \cup [M]$, and put $\epsilon(\perp) = \emptyset$, $\epsilon(a) = \{a\}$ for $a \in [M]$. The unmarked slot map on the slot and parity register is

$$\mathsf{T}_q = \sum_{a \in \{\perp\} \cup [M]} P_a^{(q)} \otimes X_{\epsilon(a)}.$$

It is unitary. At a marked slot we replace this map by the coordinate extraction

$$\mathsf{R}_q \xi = \bigoplus_{a \in [M]} P_a^{(q)} \xi.$$

The summands are mutually orthogonal, so R_q is a contraction; its omission of the vacuum summand is harmless because a marked value belongs to $Q \subseteq [M]$. The direct sum here is the canonical orthogonal decomposition of the actual slot space into the ranges of the coordinate projectors; it is not a collection of freely copied label spaces. Because all slots in one batch are parallel, the chosen cut can pass immediately after every marked extraction in that batch. Products of the extraction maps then give mutually orthogonal joint row projections P_{α}^{ket} and joint column projections P_{β}^{bra} on the physical cut spaces. These are orthogonal ranges at the cut; an outcome-dependent continuation need not preserve their orthogonality after it is pushed to the terminal boundary, which is why the intervening block contraction below cannot be removed.

At a node h , write

$$\mathsf{W}_h \xi = \bigoplus_{y \in Y_h} W_{h,y} \xi.$$

This is an isometry. The preparation map $z \mapsto z \otimes \psi_h$, the unmarked toggles, the extraction maps, the controlled no-later-occurrence projections, and the controlled terminal projections $\Pi_{\alpha}^{\text{res}}$ or Π_{β}^{res} are therefore all contractions.

For reference, put

$$P_{\mathbf{a}} = \left(\bigotimes_{q=1}^{t_h} P_{a_q}^{(q)} \right) \otimes I_{\mathcal{A}_h}.$$

When the local batch contains no marked slot, the resulting map on the incoming parity register is the explicitly defined isometry

$$J_h |S\rangle = \bigoplus_{y \in Y_h} \sum_{\mathbf{a}} |S \Delta \partial \mathbf{a}\rangle \otimes W_{h,y} P_{\mathbf{a}} |\psi_h\rangle. \quad (7)$$

Indeed, for $S, T \subseteq [M]$,

$$\begin{aligned}
\langle J_h T, J_h S \rangle &= \sum_{y, \mathbf{a}, \mathbf{b}} \mathbf{1}_{\{T \triangle \partial \mathbf{b} = S \triangle \partial \mathbf{a}\}} \langle P_{\mathbf{b}} \psi_h, M_{h,y} P_{\mathbf{a}} \psi_h \rangle \\
&= \sum_{\mathbf{a}, \mathbf{b}} \mathbf{1}_{\{T \triangle \partial \mathbf{b} = S \triangle \partial \mathbf{a}\}} \langle P_{\mathbf{b}} \psi_h, P_{\mathbf{a}} \psi_h \rangle \\
&= \mathbf{1}_{\{S=T\}}.
\end{aligned}$$

The marked local maps are obtained from (7) by replacing the appropriate toggles by the extraction maps R_q , retaining the extracted values as open orthogonal range factors, and composing the same-side controlled projections described above. The cut-edge spaces are defined recursively. At a terminal history they consist of the residual parity wire. At an internal history h , the ket cut-edge space is the direct sum, over $y \in Y_h$, of tensor products of the parent residual system $\mathcal{E}_{h,y}$, the ket cut-edge space of the child hy , and the orthogonal range factors created by any local ket extractions; the bra space is defined in the same way. We denote these spaces by $\mathcal{U}_h$ and $\mathcal{V}_h$.

Contracting the local ket tensors from the entrance of the batch to these cut edges defines

$$A_{h,k} : \mathcal{U}_h^{\text{in}} \longrightarrow \bigoplus_{y \in Y_h} \mathcal{E}_{h,y} \otimes \mathcal{U}_{hy},$$

and contracting the corresponding bra tensors defines

$$A_{h,b} : \mathcal{V}_h^{\text{in}} \longrightarrow \bigoplus_{y \in Y_h} \mathcal{E}_{h,y} \otimes \mathcal{V}_{hy}.$$

Here the harmless notation $\mathcal{U}_h^{\text{in}}, \mathcal{V}_h^{\text{in}}$ denotes the wires entering the local portion of the chosen cut; local extraction range factors are included on the appropriate side of these maps. Thus these maps are completely specified by the preparation injection, the slot maps T_q and R_q , the instrument isometry, and the controlled record projections. The construction above gives

$$\|A_{h,k}\| \leq 1, \quad \|A_{h,b}\| \leq 1.$$

Choose a zig-zag cut through the doubled acyclic network that intersects each marked ket slot immediately after its ket extraction and each marked bra slot immediately after its bra extraction. This is where the fresh architecture first becomes load bearing. The t_h query slots inside one batch are parallel, and no quantum-memory wire passes from a parent batch to a child batch, so the cut is genuinely separating and never crosses one persistent quantum wire twice. In a coherent serial-query circuit, marks at different query times lie on the same evolving wire and no analogous family of simultaneous physical cut projections exists. The cut edges are actual physical slot, environment, open-coordinate, and parity wires. Contracting all one-sided tensors exterior to these cuts, starting with the prepared states and the empty parity register, produces

$$u = \bigoplus_{\alpha} u_{\alpha}, \quad v = \bigoplus_{\beta} v_{\beta}.$$

The extraction ranges are orthogonal and every one-sided tensor is a contraction, hence

$$\|u\|^2 = \sum_{\alpha} \|u_{\alpha}\|^2 \leq 1, \quad \|v\|^2 = \sum_{\beta} \|v_{\beta}\|^2 \leq 1.$$

The tensors remaining between the two sides of the cut define one operator $E : \bigoplus_{\beta} \mathcal{V}_{\beta} \rightarrow \bigoplus_{\alpha} \mathcal{U}_{\alpha}$. The spaces $\mathcal{U}_{\alpha}$ and $\mathcal{V}_{\beta}$ are the ranges of the joint physical cut projections, not abstract copies of one common space. This operator is defined recursively, not entry by entry. At a terminal history h , it is the scalar acceptance factor a_h times the identity identification of the two residual parity registers, with the same-side residual projections already included in the side maps. At an internal history it is

$$E_h = A_{h,k}^* \left(\bigoplus_{y \in Y_h} I_{\mathcal{E}_{h,y}} \otimes E_{hy} \right) A_{h,b}. \quad (8)$$

The direct-sum decomposition records the common classical outcome. The factor $I_{\mathcal{E}_{h,y}} \otimes E_{hy}$ is the second load-bearing use of freshness: the child acts on a newly prepared system and never acts on the residual quantum system of the parent batch. Together with the separating-cut property above, this is what distinguishes the proof from a coherent serial-query calculation. Backward induction gives

$$\|E_h\| \leq 1$$

at every history. In particular $\|E\| \leq 1$, and its blocks are

$$E_{\alpha\beta} = P_{\alpha}^{\text{ket}} E P_{\beta}^{\text{bra}}.$$

For a single batch this construction is the familiar formula

$$u_{\alpha} = P_{\alpha}^{\text{ket}} \psi, \quad v_{\beta} = P_{\beta}^{\text{bra}} \psi, \quad E = \sum_y a_y M_y,$$

so the block family is obtained by compressing one positive contraction, not by pairing unrelated vectors through an entrywise identity.

Expanding the maps $A_{h,k}$, $A_{h,b}$, and (8) in the slot bases reproduces exactly the products of local matrix elements in (5). The parity identity, the no-later-occurrence projections, and the residual-disjointness projections select exactly the histories with the fixed record, as proved above. Multiplication by $C_{\alpha\beta}$ inserts the coefficient c_Q for the distinct exposed set Q . This proves (6) with the stated vectors and the stated single block contraction. $\square$

Remark 4.3 (Why the middle contraction cannot be discarded). It is tempting to push the complete measured continuation into the side vectors and replace E by the identity. That is false even for one batch. Take a two-dimensional query space with coordinate basis $|1\rangle, |2\rangle$, prepare $\psi = (|1\rangle + |2\rangle)/\sqrt{2}$, measure in the $|+\rangle, |-\rangle$ basis, and accept only the $+$ outcome. The full instrument is an isometry, but the accepted terminal vectors associated with the physical inputs $|1\rangle$ and $|2\rangle$ are both proportional to $|+\rangle$ and are not orthogonal. The correct single-batch representation keeps the orthogonal vectors $u_{\alpha} = P_{\alpha} \psi$ at the physical cut and places the outcome-dependent acceptance in the contraction $E = \sum_y a_y M_y$. The general construction in Lemma 4.2 is the adaptive version of exactly this separation.

4.3 A block Schur estimate

For a scalar matrix $C = (C_{\alpha\beta})$, define

$$\gamma_2(C) = \inf_{C_{\alpha\beta} = \langle p_{\alpha}, q_{\beta} \rangle} \left(\max_{\alpha} \|p_{\alpha}\| \right) \left(\max_{\beta} \|q_{\beta}\| \right).$$

Lemma 4.4 (Block Schur estimate). *Under the hypotheses of Lemma 4.2,*

$$\left| \sum_{\alpha, \beta} C_{\alpha\beta} \langle u_\alpha, E_{\alpha\beta} v_\beta \rangle \right| \leq \gamma_2(C).$$

If C has at most M^k rows, at most M^{r-k} columns, and $|C_{\alpha\beta}| \leq c_$, then*

$$\gamma_2(C) \leq c_* M^{\min(k, r-k)/2} \leq c_* M^{r/4}.$$

Proof. Choose a factorization $C_{\alpha\beta} = \langle p_\alpha, q_\beta \rangle$. Tensor E with the factorization Hilbert space and apply it to the direct-sum vectors

$$\bigoplus_{\alpha} p_\alpha \otimes u_\alpha, \quad \bigoplus_{\beta} q_\beta \otimes v_\beta.$$

The contraction of E , Cauchy–Schwarz, and the two energy bounds prove the first assertion.

For the second assertion, suppose first that $M^k \leq M^{r-k}$. Take the row factors to be the standard basis vectors and take the column factor for a column of C to be that column. The row norms are one and the column norms are at most $c_* M^{k/2}$. If the number of columns is smaller, transpose. $\square$

Proof of Theorem 4.1 for finite trees. Uniform averaging against χ_Q keeps exactly the doubled histories whose total coordinate parity is Q . Partition those histories by their unique last-occurrence record. A record with k ket marks and $r - k$ bra marks has contribution at most

$$M^{\min(k, r-k)/2} \max_{|Q|=r} |c_Q|$$

by Lemmas 4.2 and 4.4, and there are at most $\binom{D}{k} \binom{D}{r-k}$ such records. Summing over k gives the first inequality in (4); Vandermonde’s identity and $M^{\min(k, r-k)/2} \leq M^{r/4}$ give the second. $\square$

4.4 Standard-Borel outcomes and infinite trees

Lemma 4.5 (Finite approximation). *For the finite input domain $\{\pm 1\}^M$, every standard-Borel fresh protocol of hard dose at most D is a uniform limit of finite-outcome, finite-width fresh protocols of hard dose at most D .*

Proof. If $D = 0$, the entire protocol is input-independent and is already a finite classical kernel up to an arbitrarily accurate finite coarse-graining. Assume henceforth that $D \geq 1$.

Every positive charge is an integer. Hence a branch of dose at most D contains at most D positive-charge nodes. Collapse every zero-charge subtree into its input-independent conditional kernel over the next positive-charge history, a terminal outcome, or nontermination; regard nontermination as a terminal outcome.

Proceed backward over the remaining positive-charge depth. At a node with standard-Borel outcome y , let

$$g_y = (g_y(x))_{x \in \{\pm 1\}^M} \in [0, 1]^{2^M}$$

be the continuation acceptance vector. The map $y \mapsto g_y$ is measurable. Partition the finite-dimensional cube into finitely many Borel cells of supremum diameter at most δ . Coarse-grain the POVM by the inverse images of these cells and select one actual continuation from each nonempty cell. For every input, the change in acceptance probability at that node is at most δ . Iterating through at most D positive-charge levels with $\delta = \varepsilon/D$ gives a uniform ε approximation without increasing any branch charge. $\square$

Every Fourier coefficient is continuous under uniform convergence. Equation (4) therefore passes to standard-Borel outcomes, singular histories, unrestricted width, private randomness, and infinite zero-charge tails. Equation (3) passes to the limit as well.

5 The hard distributions

Put

$$\rho = \frac{200}{201}.$$

For $s = 1, 2, 3$, independently sample jointly Gaussian vectors

$$(Z_s, W_s) \in \mathbb{R}^N \times \mathbb{R}^N$$

with

$$\mathbb{E}Z_s Z_s^\top = I, \quad \mathbb{E}W_s W_s^\top = I, \quad \mathbb{E}Z_s W_s^\top = \rho H.$$

The block covariance is positive semidefinite because H is symmetric orthogonal and $0 < \rho < 1$. Put

$$a_s = \text{sgn } Z_s, \quad b_s = \text{sgn } W_s.$$

Choose independent uniform gauges $\zeta_1, \zeta_2, \zeta_3 \in \{\pm 1\}$, set

$$\zeta_4 = \sigma \zeta_1 \zeta_2 \zeta_3, \quad \sigma \in \{\pm 1\},$$

and define

$$\begin{aligned} x^{(1)} &= \zeta_1 a_1, \\ x^{(2)} &= \zeta_2 (b_1 \odot a_2), \\ x^{(3)} &= \zeta_3 (b_2 \odot a_3), \\ x^{(4)} &= \zeta_4 b_3. \end{aligned} \tag{9}$$

Let μ_σ be the resulting law.

6 Signed Fourier moments

For $A, B \subseteq [N]$, define

$$L(A, B) = \mathbb{E} \left[\prod_{i \in A} a_i \prod_{j \in B} b_j \right]$$

for one Gaussian link.

Lemma 6.1 (One-link moment bound). *Let $a = |A|$, $b = |B|$, and $a + b \leq r \leq \sqrt{N}$. Then*

$$|L(A, B)| \leq \left(\frac{r}{2\sqrt{N}} \right)^{\max(a, b)}.$$

Proof. If both sets are empty, the assertion is immediate. If exactly one of A, B is empty, the moment is zero. Assume $a, b > 0$. The cross-covariance matrix between Z_A and W_B is

$$R = \rho H_{A, B}, \quad \|R\| \leq \|R\|_F = \rho \sqrt{\frac{ab}{N}} \leq \frac{a+b}{2\sqrt{N}} \leq \frac{r}{2\sqrt{N}}, \quad \|R\| \leq \rho < 1.$$

The function

$$F(Z_A) = \prod_{i \in A} \text{sgn } Z_i$$

is odd in each of its a coordinates, so its multivariate Hermite expansion begins in total degree at least a . The analogous function of W_B begins in total degree at least b .

For jointly standard Gaussian vectors with cross-covariance R , the conditional-expectation operator between the two Gaussian L^2 spaces is the second quantization of R . On homogeneous chaos degree p its operator norm is at most $\|R\|^p$. If F_p, G_p are the degree- p components and $m = \max(a, b)$, then

$$\begin{aligned} |L(A, B)| &\leq \sum_{p \geq m} \|R\|^p \|F_p\|_2 \|G_p\|_2 \\ &\leq \|R\|^m \left(\sum_p \|F_p\|_2^2 \right)^{1/2} \left(\sum_p \|G_p\|_2^2 \right)^{1/2} \\ &\leq \|R\|^m, \end{aligned}$$

because F and G are sign-valued. □

Let $Q = (Q_1, Q_2, Q_3, Q_4)$, with Q_b contained in block b . Write

$$q_b = |Q_b|, \quad r = q_1 + q_2 + q_3 + q_4.$$

Lemma 6.2 (Signed Fourier moment). *If $r \leq \sqrt{N}$, then*

$$|(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-})\chi_Q| \leq 2^{1-3r/4} r^{3r/4} N^{-3r/8}. \quad (10)$$

The left-hand side is zero unless all four q_b are odd.

Proof. The gauge factor in $\chi_Q(x)$ is

$$\sigma^{q_4} \zeta_1^{q_1+q_4} \zeta_2^{q_2+q_4} \zeta_3^{q_3+q_4}.$$

Gauge averaging is therefore zero unless

$$q_1 \equiv q_2 \equiv q_3 \equiv q_4 \pmod{2}.$$

The difference between $\sigma = +1$ and $\sigma = -1$ is zero unless the common parity is odd. In the surviving case,

$$(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-})\chi_Q = 2L(Q_1, Q_2)L(Q_2, Q_3)L(Q_3, Q_4).$$

Before coarsening by the total level, Lemma 6.1 gives the composition-dependent estimate

$$|(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-})\chi_Q| \leq 2 \prod_{s=1}^3 \left(\rho \sqrt{\frac{q_s q_{s+1}}{N}} \right)^{\max(q_s, q_{s+1})}.$$

For nonnegative $q_1, \dots, q_4$,

$$\begin{aligned} \max(q_1, q_2) &\geq \frac{3}{4}q_1 + \frac{1}{4}q_2, \\ \max(q_2, q_3) &\geq \frac{1}{2}q_2 + \frac{1}{2}q_3, \\ \max(q_3, q_4) &\geq \frac{1}{4}q_3 + \frac{3}{4}q_4. \end{aligned}$$

The sum of the three right-hand sides is $3r/4$. Apply Lemma 6.1 to the three links. □

7 Average-case indistinguishability

Proposition 7.1. *If a passive protocol has dose*

$$D < \frac{2}{15} N^{1/8},$$

then its acceptance probability f satisfies

$$|\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f| < \frac{14641}{109375}.$$

Proof. By (3),

$$\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f = \sum_{r=0}^{2D} \sum_{|Q|=r} \hat{f}(Q) (\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-}) \chi_Q.$$

Only $r \geq 4$ contributes. Under the assumed dose bound, $2D < \sqrt{N}$, so Theorem 4.1 and Lemma 6.2 apply at every nonzero level:

$$\begin{aligned} |\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f| &\leq 2 \sum_{r=4}^{2D} \binom{2D}{r} (4N)^{r/4} 2^{-3r/4} r^{3r/4} N^{-3r/8} \\ &\leq 2 \sum_{r=4}^{2D} \left(2^{3/4} e D r^{-1/4} N^{-1/8} \right)^r. \end{aligned}$$

Here we used $\binom{2D}{r} \leq (2eD/r)^r$. For $r \geq 4$,

$$2^{3/4} r^{-1/4} = 2 \cdot 2^{-1/4} r^{-1/4} \leq 2 \cdot 8^{-1/4}.$$

Thus the last display is at most

$$2 \sum_{r=4}^{\infty} q^r = \frac{2q^4}{1-q}, \quad q = 2eDN^{-1/8} 8^{-1/4}.$$

If $D < 2N^{1/8}/15$, then, using $e < 11/4$ and $8^{1/4} > 5/3$,

$$q < \frac{11}{25}.$$

Consequently

$$\frac{2q^4}{1-q} < \frac{2(11/25)^4}{1-11/25} = \frac{14641}{109375}.$$

□

8 Mean and concentration of the hard distributions

8.1 Exact mean and exact normalization

The Gaussian arcsine identity gives

$$\mathbb{E}[a_{s,i} b_{s,j}] = \frac{2}{\pi} \arcsin(\rho H_{ij}) = \beta_N H_{ij},$$

where

$$\beta_N = \frac{2\sqrt{N}}{\pi} \arcsin\left(\frac{\rho}{\sqrt{N}}\right).$$

Using the exact $N^{-5/2}$ coefficient and independence of the three links,

$$\mathbb{E}_{\mu_\sigma} \text{Forr}_4(x) = \sigma \beta_N^3. \quad (11)$$

Indeed, the three link moments contribute $N^{-3/2}$, the defining coefficient contributes $N^{-5/2}$, and the four-index sum contributes N^4 .

Since $\arcsin z \geq z$, $\rho = 200/201$, and $\pi < 22/7$,

$$\beta_N > \frac{1400}{2211}.$$

Therefore

$$\beta_N^3 - \frac{1}{4} > \left(\frac{1400}{2211}\right)^3 - \frac{1}{4} = \frac{167480069}{43234079724} > \frac{1}{260}. \quad (12)$$

8.2 A conditional-independent representation

Every monomial in Forr_4 contains one coordinate from each block, and

$$\zeta_1 \zeta_2 \zeta_3 \zeta_4 = \sigma.$$

Therefore the gauges contribute the deterministic overall factor σ to Forr_4 and do not contribute to its variance. We may analyze the ungauged block vectors

$$a_1, \quad b_1 \odot a_2, \quad b_2 \odot a_3, \quad b_3$$

and restore the factor σ at the end.

Let G_s, ξ_s, η_s be independent standard Gaussian vectors and set

$$a_s = \text{sgn}\left(G_s + \frac{\xi_s}{\sqrt{200}}\right), \quad b_s = \text{sgn}\left(HG_s + \frac{\eta_s}{\sqrt{200}}\right).$$

After standardization, this pair has cross-covariance ρH , and its signs have the same law as the preceding a_s, b_s . Conditional on G_1, G_2, G_3 , the primitive signs $a_1, b_1, a_2, b_2, a_3, b_3$ are mutually independent across all coordinates; hence the $4N$ ungauged input coordinates displayed above are mutually independent. Moreover,

$$\mathbb{E}[a_{s,i}|G_s] = \phi(G_{s,i}), \quad \mathbb{E}[b_{s,j}|G_s] = \phi((HG_s)_j),$$

where

$$\phi(z) = \text{erf}(10z), \quad L := \|\phi'\|_\infty = \frac{20}{\sqrt{\pi}}, \quad L^2 < 128.$$

The last inequality follows from $\pi > 25/8$.

Put

$$M(g) = D_{\phi(g)} H D_{\phi(Hg)}.$$

Every $M(g)$ is a contraction. The conditional mean of Forr_4 is

$$\overline{F} = \sigma \langle u | M(G_1) M(G_2) M(G_3) | u \rangle. \quad (13)$$

8.3 Conditional rounding variance

Let $\text{Forr}_4^{\text{ung}}$ denote fourfold forrelation evaluated on the ungauged block vectors

$$y^{(1)} = a_1, \quad y^{(2)} = b_1 \odot a_2, \quad y^{(3)} = b_2 \odot a_3, \quad y^{(4)} = b_3.$$

Then $\text{Forr}_4 = \sigma \text{Forr}_4^{\text{ung}}$ exactly, so the two variables have the same variance. Conditional on G_1, G_2, G_3 , all coordinates of the four vectors $y^{(b)}$ are independent, but generally biased. For one independent sign Y with $\Pr(Y = 1) = p$, a multilinear function satisfies, conditional on all other coordinates,

$$\text{Var}_Y(F) = 4p(1-p)(\partial_Y F)^2 \leq (\partial_Y F)^2.$$

Here $\partial_Y F = (F|_{Y=+1} - F|_{Y=-1})/2$, so this convention agrees with the coefficient of Y in the multilinear expansion. The conditional Efron–Stein inequality therefore gives

$$\text{Var}(\text{Forr}_4^{\text{ung}} | G_1, G_2, G_3) \leq \sum_{b,i} \mathbb{E}[(\partial_{b,i} \text{Forr}_4^{\text{ung}})^2 | G_1, G_2, G_3].$$

We average this inequality over the latent Gaussians. For blocks 1 and 4, the derivative vector is $N^{-1/2}$ times a vector of norm at most one, so the sum of squared derivatives is at most $1/N$ pointwise.

For block 2, the derivative at coordinate j has the form $\ell_j r_j$, where

$$\ell = HD_{a_1} u, \quad r = HD_{b_2 \odot a_3} HD_{b_3} u.$$

The vector a_1 is independent of (b_2, a_3, b_3) , hence ℓ and r are independent. Marginally, a_1 is an i.i.d. uniform sign vector, so $\mathbb{E}\ell_j^2 = 1/N$, while $\|r\| \leq 1$. Therefore

$$\mathbb{E} \sum_j \ell_j^2 r_j^2 \leq \frac{1}{N}.$$

For block 3, the left partial-chain vector depends on (a_1, b_1, a_2) , while the right vector $HD_{b_3} u$ depends only on the independent i.i.d. uniform sign vector b_3 ; the same argument gives another $1/N$. These are unconditional calculations after the conditional Efron–Stein inequality has been averaged; no conditional uniformity is asserted. Consequently,

$$\mathbb{E} \text{Var}(\text{Forr}_4 | G_1, G_2, G_3) = \mathbb{E} \text{Var}(\text{Forr}_4^{\text{ung}} | G_1, G_2, G_3) \leq \frac{4}{N}. \quad (14)$$

8.4 Latent Gaussian variance

Gaussian Poincare gives

$$\text{Var}(\overline{F}) \leq \sum_{s=1}^3 \mathbb{E} \|\nabla_{G_s} \overline{F}\|^2.$$

For one link, write

$$\overline{F} = \langle p, D_{\phi(G)} HD_{\phi(HG)} q \rangle,$$

where p, q depend on disjoint collections of the other links, are independent of G and of each other, and have norm at most one. Define

$$A_i = p_i (HD_{\phi(HG)} q)_i, \quad B_j = q_j (HD_{\phi(G)} p)_j.$$

Because H is symmetric,

$$\nabla_G \bar{F} = D_{\phi'(G)} A + H D_{\phi'(HG)} B,$$

and therefore

$$\|\nabla_G \bar{F}\|^2 \leq 2L^2(\|A\|^2 + \|B\|^2). \quad (15)$$

We use the translation symmetries of the Sylvester matrix. For $s \in \mathbb{F}_2^d$, let

$$(P_s v)_i = v_{i+s}, \quad (D_s v)_i = (-1)^{\langle s, i \rangle} v_i.$$

Then

$$P_s H = H D_s, \quad H P_s = D_s H.$$

Since ϕ is odd,

$$M(P_s g) = P_s M(g), \quad M(D_s g) = M(g) P_s. \quad (16)$$

The standard Gaussian law is invariant under every P_s, D_s . It follows from (16) that every nonempty left or right partial-chain vector in (13) has a distribution invariant under the transitive translation group $\{P_s\}$. Its coordinate second moments are equal and, because its norm is at most one, are at most $1/N$.

At each link, p is either u or such a coordinate-transitive partial-chain vector; the same holds for q . The vector p is independent of G, q , and q is independent of G, p . Therefore, writing $v = H D_{\phi(HG)} q$,

$$\mathbb{E} \|A\|^2 = \sum_i \mathbb{E}[|p_i|^2 |v_i|^2] \leq \frac{1}{N} \mathbb{E} \|v\|^2 \leq \frac{1}{N}.$$

The same calculation, with $w = H D_{\phi(G)} p$, gives

$$\mathbb{E} \|B\|^2 \leq \frac{1}{N}.$$

Equation (15) gives

$$\mathbb{E} \|\nabla_G \bar{F}\|^2 \leq \frac{4L^2}{N}.$$

Summing the three links,

$$\text{Var}(\bar{F}) \leq \frac{12L^2}{N}. \quad (17)$$

By total variance, (14), (17), and $L^2 < 128$,

$$\text{Var}_{\mu_\sigma}(\text{Forr}_4) \leq \frac{4 + 12L^2}{N} < \frac{1540}{N}. \quad (18)$$

9 Transfer to the exact promises

Equations (12) and (18), followed by Chebyshev, give

$$\mu_\sigma \left(\sigma \text{Forr}_4 < \frac{1}{4} \right) \leq \frac{1540 \cdot 260^2}{N} = \frac{104104000}{N}.$$

For $N \geq 2^{30}$, put

$$\varepsilon_0 = \frac{104104000}{2^{30}} = \frac{1626625}{16777216}.$$

Define

$$\nu_\sigma = \mu_\sigma|_{\{\sigma \text{ Forr}_4 \geq 1/4\}}.$$

Then

$$\nu_+(\Pi_+) = 1, \quad \nu_-(\Pi_-) = 1, \quad \text{TV}(\mu_\sigma, \nu_\sigma) \leq \varepsilon_0.$$

Let $f(x)$ be the probability that the protocol outputs $+$. Worst-case success at least $2/3$ implies

$$\mathbb{E}_{\nu_+} f \geq \frac{2}{3}, \quad \mathbb{E}_{\nu_-} f \leq \frac{1}{3}.$$

Since $0 \leq f \leq 1$,

$$\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f \geq \frac{1}{3} - 2\varepsilon_0 = \frac{3508733}{25165824}. \quad (19)$$

On the other hand,

$$\frac{3508733}{25165824} - \frac{14641}{109375} = \frac{15314842691}{2752512000000} > 0.$$

If $D < \frac{2}{15}N^{1/8}$, Proposition 7.1 therefore contradicts (19). Hence

$$D \geq \frac{2}{15}N^{1/8}.$$

This proves Theorem 1.1.

10 Full-model audit

The preceding proof applies to the full model in the problem statement.

- Arbitrary mixed inputs are purified within the same maximum-photon support. Arbitrary bypass idlers remain in the ancilla factors.
- The slot embedding is exact for maximum sign-mode photon number on support. Repeated occupation, bosonic symmetrization, number superpositions, and vacuum coherence are all retained.
- Each batch contains one direct-sum sign-mask interaction. No known Hadamard or other processing is inserted between the four masks inside a batch.
- Only a classical outcome controls the next fresh state. Freshness is used in two coupled parts of Lemma 4.2: the marked parallel slots admit one genuine separating cut, and every continuation acts on a new tensor factor, giving the factor $I_{\mathcal{E}_{h,y}} \otimes E_{hy}$ in (8). Neither the cut projections nor this factor may be replaced by a claim that the fully propagated terminal vectors remain orthogonal; Remark 4.3 shows why.
- All outcome weights are raw outcome weights. Heraldng and postselection never divide out reach probability and never discount dose.
- The last-occurrence partition is exact: after omitting each marked toggle, the same-side residual parity is projected to be disjoint from the exposed coordinates. Together with residual-parity equality, this forces the marked side to be the odd side and prevents duplicate records.
- Finite approximation covers arbitrary finite or standard-Borel outcome spaces, singular histories, and unrestricted width.

- Every positive traversal is counted in the branchwise dose. Because positive charges are integers, a finite-dose branch has at most D positive-charge nodes. Zero-charge tails are input-independent conditional on the current classical history and are absorbed into a terminal kernel, with nontermination treated as a terminal outcome.
- Private classical randomness is included as zero-charge classical branching.
- The hard distribution uses the exact normalized Hadamard matrix and the exact coefficient $N^{-5/2}$. The transfer conditions on the exact thresholds $+1/4$ and $-1/4$.
- The last step uses worst-case success on every promised input, not an average-case correctness assumption.
- The conclusion is a branchwise hard-dose lower bound. It is not a mean-dose bound, does not cover a quantum memory carried between batches, and does not permit coherent inter-mask processing inside one fresh batch.

A Standard Gaussian facts used above

For completeness, we record the three standard Gaussian facts invoked in the proof.

A.1 Arcsine identity

If X, Y are standard real Gaussians of correlation r , rotational symmetry in the plane gives

$$\mathbb{E}[\operatorname{sgn} X \operatorname{sgn} Y] = \frac{2}{\pi} \arcsin r.$$

A.2 Second quantization

Let $X \in \mathbb{R}^a$, $Y \in \mathbb{R}^b$ be jointly standard Gaussian with cross-covariance R . After singular-value decomposing R and making orthogonal changes of variables, the normalized multivariate Hermite basis diagonalizes the conditional-expectation operator $g(Y) \mapsto \mathbb{E}[g(Y)|X]$. On homogeneous chaos degree p , every singular value is a product of p singular values of R , so the operator norm is at most $\|R\|^p$.

A.3 Gaussian Poincare

For a locally Lipschitz function F of a standard Gaussian vector,

$$\operatorname{Var}(F) \leq \mathbb{E} \|\nabla F\|^2.$$

The function in (13) is smooth and bounded, so the inequality applies directly.